

Reasoning and Decision Making: Reflective Analysis of a Decision and a Problem

Introduction

Decision making and problem solving are unavoidable parts of everyday life. Although people may feel that they are making decisions rationally, the processes involved can be influenced by limited information, assumptions, emotions and cognitive biases. Critical thinking provides a means of improving these processes by making reasoning more deliberate and by encouraging the evaluation of evidence and alternative possibilities (Halpern, 2003). This essay reflects on two separate experiences from my life. The first concerns my decision to undertake a university degree in psychology. The second concerns my attempt to improve my performance following a disappointing university assessment result. I first describe how I approached each situation at the time, without applying psychological theory retrospectively. I then reconsider each situation using concepts from the module and evaluate whether a more systematic approach could have produced a better outcome.

Decision: Choosing to Study Psychology

How I made the decision at the time

An important decision I made was to undertake a university degree in psychology. Before making the decision, I had an interest in psychology and was increasingly interested in understanding human behaviour, thinking and decision making. However, committing to a degree represented a substantial investment of time and effort. I therefore did not make the decision immediately.

Initially, I considered whether university was worthwhile for me personally. I thought about the advantages of gaining a formal qualification, developing knowledge and improving my future career opportunities. I also considered the disadvantages, particularly the amount of work involved and whether I would be able to maintain the commitment over a long period.

I spent time considering different possibilities and ultimately decided that psychology was sufficiently interesting and useful to justify the commitment. I enrolled and began studying. Once I had started, the decision became less about whether university was worthwhile and more about maintaining the commitment and completing the degree. At the time, my approach was

therefore largely based on considering what seemed important to me, thinking about the likely consequences and choosing the option that felt like the best overall fit.

How I would approach the decision now

With the knowledge gained from this module, I would approach the same decision more systematically. Halpern (2003) argues that critical thinking involves using cognitive skills and strategies that increase the likelihood of a desirable outcome. Importantly, this involves more than simply possessing reasoning skills; the thinker must recognise when those skills are needed and monitor whether they are working effectively.

The first change would be to define the decision more precisely. Rather than asking, "Should I go to university?", I would formulate the question as: **"Which educational option is most likely to support my long-term personal, educational and career goals, given my available time, resources and interests?"** This would prevent the decision from being framed too narrowly.

I would then generate realistic alternatives. These might include studying psychology, studying a different subject, delaying university, undertaking another form of professional training, or entering employment without immediately pursuing higher education. Generating alternatives is important because considering only the most obvious option can create a false sense that there is little choice.

I would next establish explicit criteria against which each alternative could be assessed. For example, I could score each option for personal interest, career usefulness, financial cost, flexibility, workload, transferable skills and long-term opportunities. This would make my reasoning more transparent rather than allowing the strongest immediate impression to dominate.

I would also actively seek disconfirming evidence. Kahneman (2011) distinguishes between rapid, intuitive thinking and slower, more deliberate thinking. My original decision contained elements of intuitive judgement because I already felt positively towards psychology. Although that feeling was relevant, it could have encouraged confirmation bias: I might have paid greater attention to information supporting university study while giving insufficient attention to disadvantages. I would therefore deliberately ask, "What evidence would suggest that this is a poor choice?" and investigate those possibilities.

The final stage would involve reviewing the decision before acting and then evaluating the outcome afterwards. This reflects Halpern's emphasis on metacognitive monitoring. I would not expect the process to guarantee a perfect decision because decisions are made under uncertainty. Instead, the aim would be to make the reasoning process more explicit and increase the probability of a satisfactory outcome.

Reflection on the two approaches

The main difference between my original and module-informed approaches is that the original was relatively intuitive and unstructured, whereas the revised approach would make the alternatives, criteria, evidence and uncertainties explicit. The original approach was not necessarily irrational. In fact, Klein's (1998) research on naturalistic decision making demonstrates that intuition can be useful, particularly when people have relevant experience. However, intuition can also be misleading when a decision is unfamiliar or emotionally important.

The module-informed approach would therefore not completely replace intuition with analysis. Instead, I would use my initial judgement as a starting point and then subject it to deliberate evaluation. This is a significant improvement because it preserves the practical value of intuition while creating opportunities to identify unsupported assumptions and alternative explanations. I believe this would have made my original decision more robust, although it would not have removed uncertainty or guaranteed a better outcome.

Problem: Responding to a Poor University Assessment Result

How I approached the problem at the time

A significant problem I encountered was receiving a disappointing result on a university assignment. I received a mark of 32, which was considerably below the standard I wanted to achieve and meant that I needed to address the weaknesses in my work.

My first reaction was disappointment because I had invested substantial time in the assignment. However, I knew that simply being dissatisfied with the mark would not change the situation. I therefore went back through the assignment and considered the feedback I had received.

I compared what I had submitted with what the assessment required. I identified areas where my response had not sufficiently addressed the question and where the quality of the argument and supporting evidence could be improved. I then planned changes to the structure and content. Instead of simply correcting individual sentences, I tried to improve the assignment as a whole.

I subsequently rewrote sections, strengthened the evidence supporting my points and attempted to make the argument more directly relevant to the question. I also paid greater attention to academic structure and referencing. The problem was therefore approached through a process of identifying weaknesses, making changes and producing an improved version rather than simply repeating the original approach.

How I would solve the problem now

Using knowledge from the module, I would first define the problem much more precisely. Newell and Simon's (1971) concept of a "problem space" is useful here because problem solving

involves moving from an initial state towards a desired goal through possible intermediate states. My initial state was an assignment that had received a low mark. My goal state was not simply "get a higher mark"; it was to produce work that fulfilled the assessment criteria and demonstrated stronger reasoning.

I would therefore divide the problem into smaller differences between the current and desired states. For example:

Current state: weak assessment performance.

Desired state: assessment demonstrating accurate knowledge, effective application, critical analysis, appropriate evidence and clear academic presentation.

I would then identify specific obstacles preventing movement between the two states. These could include misunderstanding the question, insufficiently developed arguments, weak application of theory, inadequate evidence, poor structure or ineffective referencing.

This approach resembles means-end analysis, in which the problem solver identifies differences between the current and goal states and selects actions that reduce those differences (Newell & Simon, 1972). Rather than simply deciding that the assignment "needs to be better", I would create specific subgoals. For example, if critical analysis were weak, the subgoal would be to identify the central claims in each section, evaluate the evidence supporting them, consider alternative viewpoints and reach a justified judgement.

I would also treat my initial explanation for the poor mark as a hypothesis rather than a fact. For example, I might initially believe that the main problem was insufficient academic sources. I would test this by comparing the feedback with the marking criteria. If the feedback instead showed that the central problem was failure to answer the question directly, adding more sources would not solve the real problem. This illustrates the value of hypothesis testing and avoiding premature conclusions.

Metacognitive monitoring would provide another important improvement. Rather than waiting until the final draft to judge its quality, I would monitor my progress throughout the process. After each major section, I could ask: "Does this directly answer the question?", "What evidence supports this claim?", "Have I considered an alternative interpretation?", and "Am I explaining the significance of the evidence rather than merely describing it?" This would turn evaluation into a continuous process rather than a final check.

I would also use feedback more objectively. A low mark can produce an emotional reaction, potentially encouraging either defensiveness or excessive self-criticism. A critical-thinking approach would instead treat feedback as information about the gap between the current and desired states. The objective would not be to prove that the original work was good enough, but to identify what evidence indicates that improvement is required.

Finally, I would evaluate the revised work against the same criteria used to define the problem. This creates a feedback loop: identify the problem, generate actions, implement them, evaluate

the outcome and, if necessary, revise the solution. This is preferable to assuming that completing a rewrite automatically means that the problem has been solved.

Reflection on the two approaches

The main difference between my original problem-solving approach and the module-informed approach is the level of structure. Originally, I recognised that something was wrong and responded by reviewing and rewriting the assignment. This was useful, but it relied considerably on my own judgement about what needed changing.

The module-informed approach would make the problem representation much more explicit. Instead of treating the low mark as one large problem, I would break it into component problems and establish measurable subgoals. This is consistent with Newell and Simon's (1971, 1972) account of problem solving, where the representation of the problem determines the possible routes towards a solution.

However, systematic problem solving also has limitations. Real-world problems are rarely perfectly defined, and an academic assessment cannot be reduced entirely to a mechanical sequence of steps. The same evidence can sometimes support different interpretations, and marking involves human judgement. Consequently, a highly structured approach could create unnecessary rigidity. Expertise and intuition still have a role, particularly when deciding which issues deserve the most attention.

Nevertheless, I believe the module-informed approach would probably have produced a better solution. It would have reduced the likelihood of solving the wrong problem, encouraged me to test my assumptions and provided clearer criteria for judging whether my revisions were actually improvements.

Conclusion

Reflecting on these two experiences demonstrates that effective reasoning is not simply about reaching an outcome but about improving the process used to reach it. My original decision to study psychology was largely based on personal judgement, consideration of consequences and an overall sense that it was the right choice. My response to a poor assessment result involved recognising weaknesses and attempting to correct them, but it was initially less systematic than it could have been.

The module-informed approaches introduce greater structure without suggesting that intuition should be eliminated. For the decision, I would generate alternatives, establish explicit criteria, seek disconfirming evidence and monitor my reasoning. For the problem, I would define the initial and goal states, identify specific differences, create subgoals, test hypotheses and continually evaluate progress. These approaches would not guarantee perfect decisions or solutions because uncertainty and human judgement remain unavoidable. However, they would make my reasoning more transparent, reduce the influence of unsupported assumptions and

increase the likelihood of achieving desirable outcomes. This is consistent with Halpern's (2003) view that critical thinking is ultimately concerned with using appropriate cognitive strategies to make desirable outcomes more likely.

References

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